WEEKLY LESSON PLAN · 2026-27

# Principles of Applied Engineering

TEKS §127.781 · CTE Level 1 · Grade 9 · Mr. Gavaldon · Da Vinci School
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SEMESTER 1
**Week 2**

## MONDAY Day 6 - Build-a-Car: Plan & Design

Aug 17

**OBJECTIVE**
Students will define the roll-far problem with measurable success criteria and constraints, generate and score two car concepts in a weighted decision matrix, and predict roll distance from an energy model (PE = mgh) - completing an engineering planning packet before any building.

**TEKS**
§127.781(d)(2)(G) engineering design process — planning; (d)(3)(C) sketching the design

**ELPS / EB SUPPORT**
Design brief and planning packet give sentence frames plus a labeled force diagram (VC, DP, CD). Pre-teach constraint / criteria / rolling-resistance from a picture bank (PV). Mixed teams with builder / recorder / tester roles; native-language partner talk before writing the justification (PN, WT).

**AGENDA · 45 MIN**
- 5′ Bell-work: list every factor that could make one car roll farther than another
- 10′ Design brief: the 3-day mission, constraints vs. criteria, the decision matrix
- 12′ Concept: energy (PE to KE) and why cars stop - rolling resistance & alignment
- 6′ Worked example: predict a car's roll distance from height
- 9′ You do: generate 2 concepts, score the matrix, predict distance
- 3′ Exit ticket: constraint vs. criterion

**SLIDES / ACTIVITY**
Engineering a Car That Rolls: Plan & Design

**NOTEBOOK**
Glue in the car planning sheet. Sketch your design and label the body, axles, and wheels.

**TURN IN**
Engineering Design Packet - Roll-Far Car

## TUESDAY Day 7 - Build Day

Aug 18

**OBJECTIVE**
Students will fabricate their prototype car from an approved plan using safe tool practice, achieve a running clearance fit at the wheel-axle joint, and verify axle spacing and parallelism to a stated tolerance (60.0 +/- 1.0 mm), logging every measurement and corrective action.

**TEKS**
§127.781(d)(2)(G) build the prototype (design process); (d)(4) safe, responsible use of tools & materials

**ELPS / EB SUPPORT**
Teacher models each assembly step at the front and leaves an annotated photo sequence up (VC, DP, RS). Build-log terms (tolerance, clearance, parallel) pre-taught with a picture bank (PV, CD). Roles keep every student hands-on; partner talk before recording corrective action (PN, WT).

**AGENDA · 45 MIN**
- 5′ Bell-work: name two build risks and the one dimension that must be in tolerance
- 7′ Safety protocol + fit & tolerance mini-lesson
- 6′ Teacher models axle/wheel assembly and how to measure it
- 22′ Build from approved plans: assemble, measure, roll-test as you go
- 5′ Dimensional inspection + park it + log one fix

**SLIDES / ACTIVITY**
Build Day: Precision, Fit, and Safe Fabrication

**NOTEBOOK**
Log your build: what went well, what didn't, and the one fix you'll make on test day.

**TURN IN**
Build Log & Dimensional Inspection

WEDNESDAY

## Day 8 - Test, Measure, Improve

Aug 19

**OBJECTIVE**

Students will design a fair test by controlling variables, collect multiple-trial distance data, compute mean, range, and percent improvement, visualize before/after results in a bar chart, and justify their next iteration with the data.

**TEKS**

§127.781(d)(2)(G) test & evaluate (design process); (d)(3)(D) present results with visualization

**ELPS / EB SUPPORT**

Data sheet is mostly numbers with one sentence of reasoning (low language load, CD, DP). Pre-teach independent/dependent/controlled variable and mean/range with a picture bank (PV). Pair-share the analysis before writing the conclusion (WT, PN).

**AGENDA · 45 MIN**

- 5′ Bell-work: what makes a test fair? name the three kinds of variables
- 8′ Fair testing, variables, and why we run multiple trials (mean & range)
- 6′ Worked example: mean, range, and percent improvement
- 18′ Ramp test: 3 baseline trials -> one controlled change -> 3 trials
- 5′ Analyze: compute percent change and graph before/after
- 3′ Exit ticket: which change to make next and why

**SLIDES / ACTIVITY**

Test, Measure, Improve: Data-Driven Iteration

**NOTEBOOK**

Record both trial distances and the single change you made. Was your prediction right?

**TURN IN**

Test Data & Iteration Analysis

THURSDAY

## Day 9 - The Real Tools (video + emails)

Aug 20

**OBJECTIVE**

Students will explain what CAD and parametric modeling are and why engineers design in software before building, analyze CAD vs. paper prototyping against engineering criteria, apply professional-online-conduct standards, and confirm a working school email for account creation.

**TEKS**

§127.781(d)(3)(C) computer-aided design & drafting (CADD); (d)(1)(A) professional online conduct

**ELPS / EB SUPPORT**

Video has a guided sheet with sentence stems and target terms to catch (RS, CD, DP). Pre-teach CAD / parametric / extrude with labeled screenshots (PV, VC). 'Show me your email' is a low-stakes one-on-one check with wait time (WT); native-language partner support at the station (PN).

**AGENDA · 45 MIN**

- 5′ Bell-work: what is CAD, and why not just build every idea by hand?
- 12′ What is CAD? Parametric modeling and the sketch-to-feature workflow
- 10′ Watch: first look at Fusion 360 (guided viewing sheet)
- 8′ Why CAD beats paper + the certification path + professional online conduct
- 7′ Email-confirm station + CAD-vs-paper decision matrix
- 3′ Exit ticket: define parametric

**SLIDES / ACTIVITY**

From Paper to Parametric CAD

**NOTEBOOK**

Note 3 things Fusion can do, and one thing you want to build in it this year.

**TURN IN**

CAD Concepts, Guided Viewing & Account Readiness

FRIDAY

## Day 10 - Create Account & Launch Fusion

Aug 21

**OBJECTIVE**

Students will create an Autodesk education account, launch and navigate the Fusion 360 interface, model a first parametric part (a dimensioned, extruded 60 x 30 x 3 mm chassis plate), save it to the cloud, and support a peer using a structured help protocol.

**TEKS**

§127.781(d)(3)(C) set up & navigate CAD (CADD); (d)(1)(B) collaborate & support peers

**ELPS / EB SUPPORT**

Account and modeling steps are numbered and screenshotted in Classroom (VC, DP, CD). Pre-teach interface terms (sketch, extrude, ViewCube, timeline) with labeled images (PV). Finishers pair with peers using an 'ask three, then me' protocol; native-language walkthrough allowed (PN, WT).

**AGENDA · 45 MIN**

- **5′** Bell-work: why an education license and a school email for this account?
- **7′** Account steps + tour of the Fusion interface (toolbar, browser, timeline, ViewCube)
- **8′** Demo: sketch -> dimension -> extrude a 60 x 30 x 3 mm chassis plate
- **20′** Create/verify account, launch Fusion, model your first part, save to cloud
- **5′** Peer support + first-look checklist + exit

**SLIDES / ACTIVITY**

Launch Fusion 360: Your First Parametric Part

**NOTEBOOK**

Paste your Fusion account email (not password). Note where the Sketch button and timeline are.

**TURN IN**

Fusion 360 First Session - Setup, Build Log & Peer Support